Terrain texturing

Release version 0.1 (2007-11-05)

A word about the examples in this/these documents.

We used the modelling package 3D Studio max to create Rail Simulator, so this product appears in the examples we have provided. You can, of course, use other packages such as Blender or 3D Canvas. The package we used for authoring textures is Adobe Photoshop, again you may use one of the many other packages available. Regardless of the packages you use, the document should be used as a general guide to the processes to follow to build assets for Rail Simulator

Important note: For your chosen package, you may need additional plug-ins to export the assets into Rail Simulator. We are in the process of finalising these tools and so plug-ins for some packages may not be available immediately.

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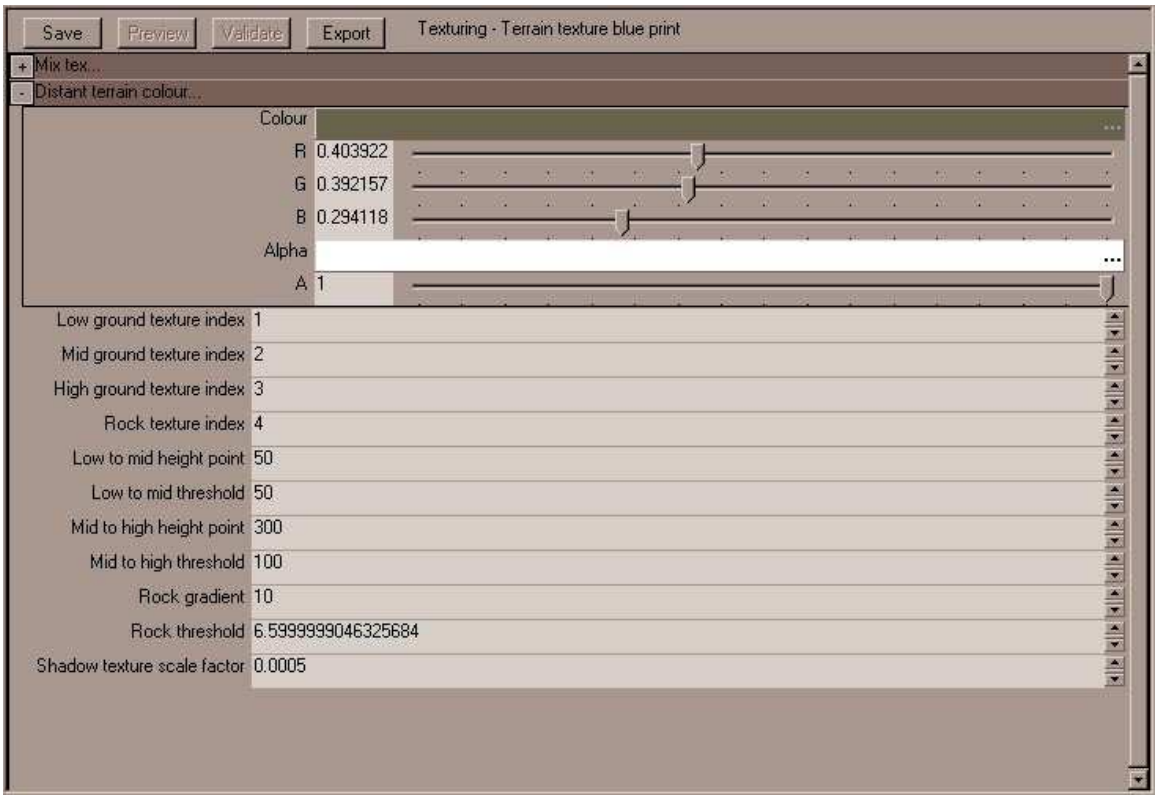
1 Texturing XML

The texture XML is located at; RailSimulator\Environment\Terrain



- **Colour:** This is the colour that the terrain texture blends after it's smallest mip, move the RGB sliders to get the desired colour.
- **Alpha:** Not used, but should be kept to 1- White.
- **Low ground texture index:** This number corresponds to the number in the mix tex list. (For example green grass)
- **Mid ground texture index:** This number corresponds to the number in the mix tex list. (For example Heather)
- **High ground texture index:** This number corresponds to the number in the mix tex list. (For example Snow, on mountain peaks)
- **Rock texture index:** This number corresponds to the number in the mix tex list. It is the texture that is displayed once the gradient is greater than the Rock gradient value.
- **Low to mid height point**: The value in metres that the low ground texture is displayed.
- **Low to mid threshold:** The value in metres in which the low and mid ground textures get mixed.

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- **Mid to high height point:** The value in metres that the low ground texture is displayed.
- **Mid to high threshold:** The value in metres in which the low and mid ground textures get mixed.
- **Rock gradient:** The height value in metres which dictates the gradient. The horizontal number is 8m (the grid in Rail Simulator is 8/8): For example, 10m high every 8m.
- **Rock Threshold:**
- **Shadow texture scale factor:** Not used in current version

1.1 Mix Tex

- **Base texture name:** Folder location and name of texture without the wang suffix. (_A/_B/_C/_D/_E/_F/_G/_H)
- **Base spring texture name:** Folder location and name of texture without the wang suffix but including the spring suffix _sp)
- **Base summer texture name** Folder location and name of texture without the wang suffix but including the summer suffix _su)
- **Base Autumn texture name** Folder location and name of texture without the wang suffix but including the autumn suffix _au)
- **Base Winter texture name** Folder location and name of texture without the wang suffix but including the winter suffix _wi)
- **Display name:** in game name.

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- **Localised display name:**
- **Wang**: If using wang textures then set this to true, if standard tile set to false and it will not look for the wang suffix.
- **Flora index:** This number represents mapping coordinates in the grass pack .dds
- **Flora density**: This number is set between 0 and 1, 1 being the most dense and 0 being sparse.
- **Category:** Pull down menu texture description.

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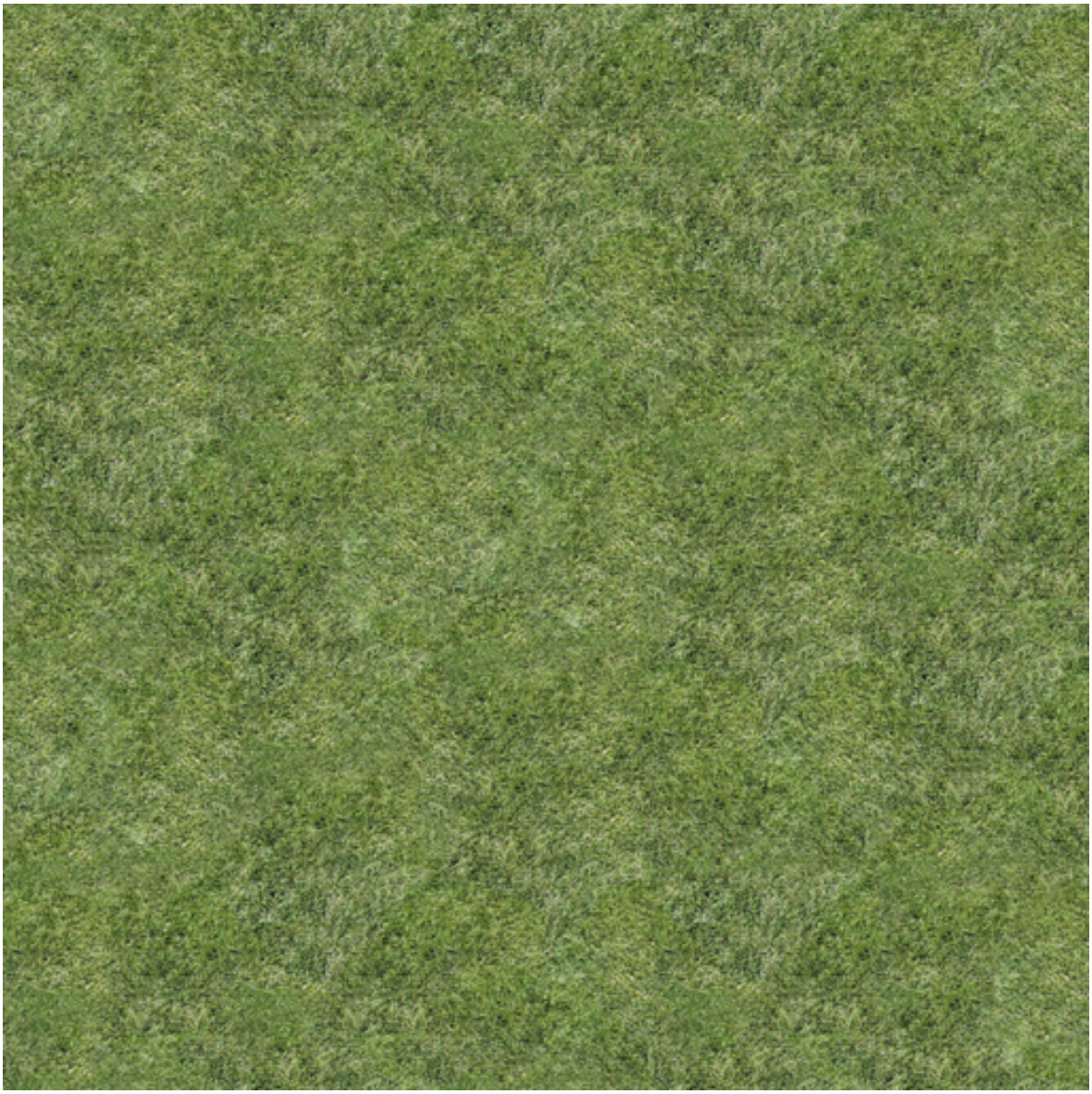
2 Wang tiles

Wang tiles are sets of tiles that fit together to non-periodically tile across the terrain, creating expanses of non repeating texture. (See example grass tiles below)



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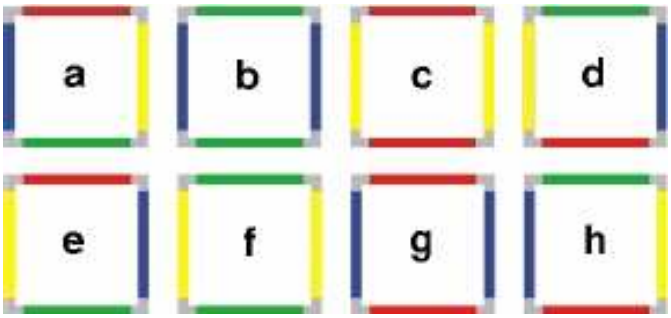
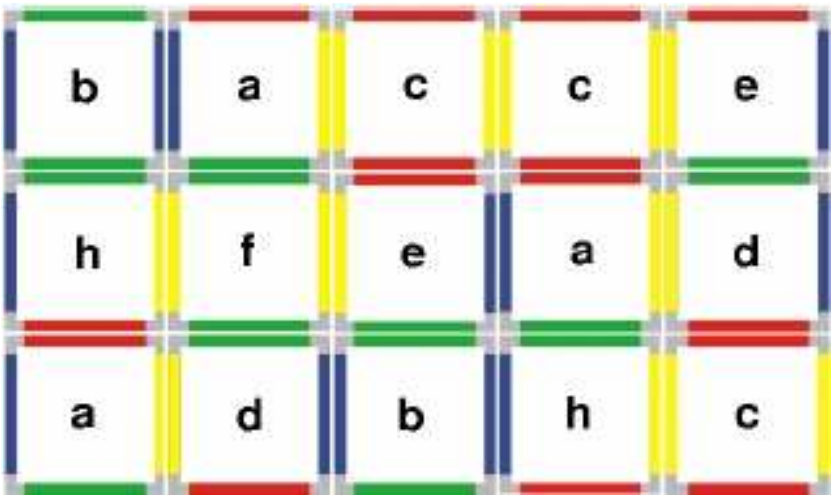
2.1 How it Works

Each edge of a Wang tile is assigned a colour.



The tiles are laid on a plane so that the colours of adjacent edges match.



A minimum of 8 tiles can be used to generate non-repeating textures. The more tiles you have obviously the less chance of repartition and tiling.

Horizontal and vertical edges each have 2 colour possibilities

There is exist 4 (2^2) possible North-West edge combinations.

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2.2 Creating Wang Tiles

Four tiles were cut from the source image; these will be used to create the 8 Wang tiles

Blue

Green

Red

Yellow

Each tile is allocated a colour and the pieces cut will be used to create the tiles

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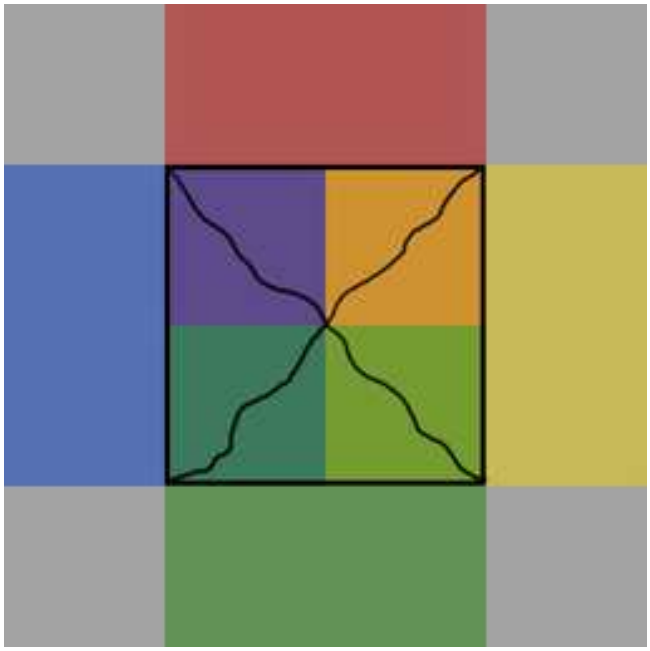
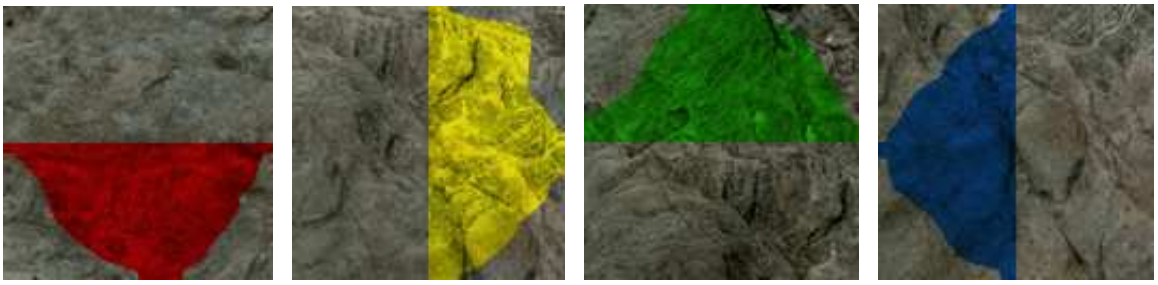


For example to create tile (e) the following pieces are cut from the initial 4 tiles



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2.3 File format

Each terrain texture needs an alpha channel. The game uses this information for blending. The darker the shade of grey the quicker the texture fades into another.

The terrain textures need to be 256/256 pixels square and saved out as a 32 bit .ace file(V2). Before saving make sure the 'Make mips' check box is ticked.

Wang textures need to be named XXXX_A.ace to XXXX_H.ace

Winter versions (For more information on season suffix see document Asset authoring guidelines) of terrain textures need to be named XXXX_wi_A.ace to XXXX_wi_H.ace

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Standard tiles

The standard tile is used in situations where a repeating pattern is desirable.

Example of standard tile terrain name;

Ploughed000_Su.ace
Ploughed000_Au.ace
Ploughed000_Wi.ace
Ploughed000_Sp.ace

Or if the texture does not have a season variant:

Ploughed000.ace (default texture that will be used in all seasons)

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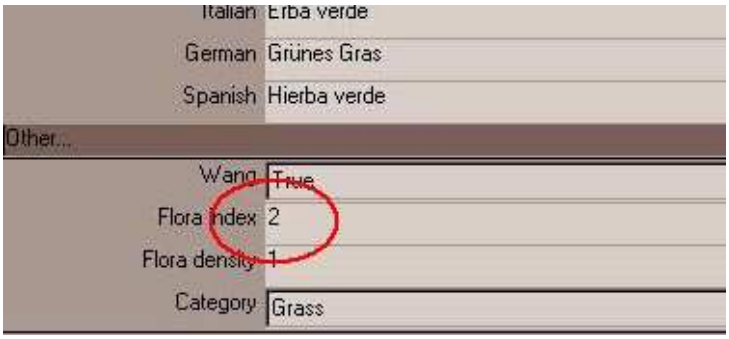
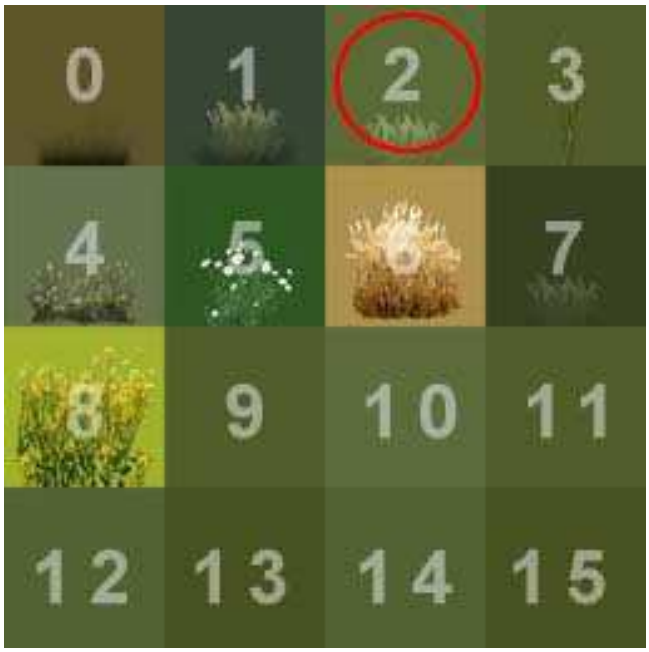
4 Procedural flora

Procedural flora can be linked to the terrain textures.
The flora index number corresponds to a tile on the grasspack.dds



The scrub texture page is called grasspack.dds. The texture page (1024x1024) consists of a bank of 16 smaller 256x256 textures arranged in a grid. See below



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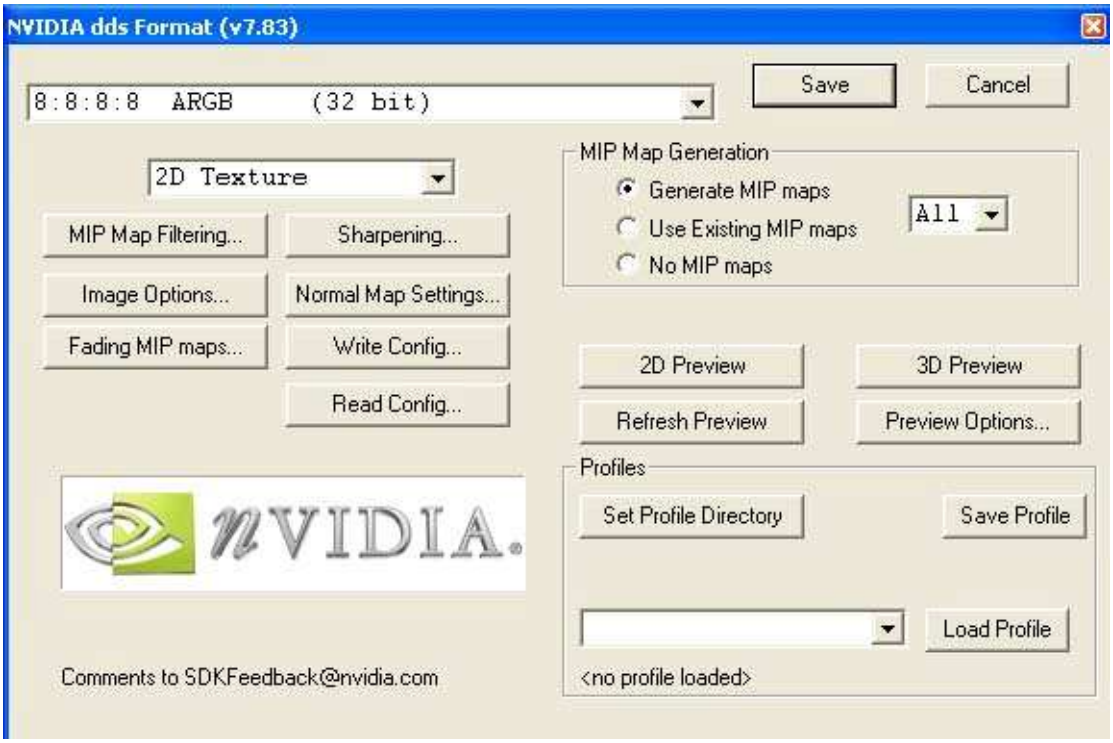
To create the .dds file the plugin from the official NVidia site was used with the values seen below.



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